Code No: 123BP **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, September/October - 2023
DATA STRUCTURES
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	How do you measure the algorithm running time?	[2]
b)	Describe best case, average case and worst case efficiency of an algorithm.	[3]
c)	List the applications of queue.	[2]
d)	Define Circular Queue full condition.	[3]
e)	State the properties of a Binary Tree.	[2]
f)	Differentiate Max-heap and Min-heap.	[3]
g)	What is searching and list the types of searching techniques.	[2]
h)	Write the applications of binary search.	[3]
i)	Define binary search tree with example.	[2]
j)	Discuss the drawbacks of AVL trees.	[3]

PART - B

(50 Marks)

2. Explain Array and Linked representation of Sparse Matrix. [10]

OR

3.a) Explain how to create circular linked list and insert nodes at end.
b) Write algorithm for above operations. [5+5]

4. Explain the procedure to convert infix expression into postfix expression and write algorithm for it. [10]

OR

5. Write C programs to implement a double ended queue ADT using arrays. [10]

6.a) Explain tree traversals with example.
b) Explain different ways of representation of graphs. [5+5]

OR

7.a) Explain with an example how to insert an element to maxheap.
b) Given In order traversal of a binary tree is D,G,B,E,A,H,F,I,C and pre order traversal is A,B,D,G,E,C,F,H,I construct binary tree. [5+5]

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8.a) Explain linear search with example.
b) Compare and contrast various sorting techniques. [3+7]

OR

9.a) State and explain heap sort with an example.
b) Apply radix sort on the following list of elements 45, 37,05,09,06,11,18,27. [5+5]

10.a) Construct a binary search tree for the following 80, 40, 75, 30, 20, 90, 50.
b) What is a trie? Briefly describe compressed tries with its disadvantages. [5+5]

OR

11.a) Construct an AVL Tree for following elements:10,20,15,3,2,16,18,26.
b) Explain about the Knuth-Morris-Pratt pattern matching algorithm. [5+5]

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